

Cédric Emmanuel KIRE

Paris, France · github.com/cedrickirek

Available from September 2026

PROFILE

Strong interdisciplinary background in maths, engineering, academic research, and business analytics. I have been trained across 4 countries (England, Japan, Cote d'Ivoire, France) where I developed an advanced self-learning mindset, a collaborative attitude, attention to detail, and a proven ability to be flexible to go beyond expectations.

EDUCATION

ÉCOLE POLYTECHNIQUE (L'X) – HEC PARIS, France September 2024 – August 2026
MSc Data Science & AI for Business — joint double degree

- Coursework: **Machine & Deep Learning, LLMs, Statistics, Optimization, Reinforcement Learning, Interpretability of ML models, Introduction to Finance**
- Research thesis — **LLM evaluation**: designed and ran an offline evaluation experiment for an LLM-as-a-Judge system scoring solutions to advanced mathematics high school questions. Built a human ground-truth benchmark from the annotations of 10 student annotators, then measured judge–human agreement to validate automated scoring against manual grading. Also built a curriculum-grounded RAG pipeline for personalised tutoring.

TOKYO INSTITUTE OF TECHNOLOGY, Tokyo, Japan April 2022 – August 2024
MEng Global Engineering & Management — Grade: 94.6/100

- Statistics, Data Analysis in Energy Markets, Energy Economics, Sustainable Finance, Project Management

UNIVERSITY OF WOLVERHAMPTON, England September 2018 – June 2021
BEng Chemical Engineering — First Class Honours

PROFESSIONAL & RESEARCH EXPERIENCE

L'ORÉAL — Data Scientist / Analyst Intern, Recommender Systems March 2025 – August 2025
Clichy, France

- Designed a **BigQuery pipeline** adding **20M+ customer records** to the recommender system's dataset, using regex-based text normalisation to recover records previously unusable due to malformed fields.
- Built a **gender-classification model** on a **90/10 imbalanced dataset**, reaching **75% recall at 70% precision**; selected the decision threshold from precision–recall trade-offs on a held-out set rather than optimising accuracy, which is uninformative at that class balance.
- Researched **state-of-the-art e-commerce recommendation algorithms** against the in-house system through a structured literature review, and proposed structural changes to the ranking approach.
- Delivered a **6-page Looker dashboard** analysing Yves Saint Laurent's 2022–2024 market performance in the UK and France, sourced from L'Oréal's enterprise data warehouse.

TOKYO INSTITUTE OF TECHNOLOGY — Research Student in Applied Machine Learning Sept 2022 – July 2024
Yokohama, Japan

- Conducted research in a **20+ member lab** (PhD and Master's students) led by Prof. Fumitake Takahashi.
- Built **regression models from scratch** to predict flood-generated waste volumes in Abidjan, achieving predictive performance **2x better than comparable published studies**.

KOZO KEIKAKU ENGINEERING — Data Scientist Consultant Intern September 2023
Tokyo, Japan

- Built **XGBoost and Random Forest** models to predict design requirements for friction materials used in the client's industrial machinery.

ENTREPRENEURSHIP

Start-up Launchpad, HEC Paris — Tutorly: hybrid Human–AI tutoring system 2026

- Building an AI-powered tutoring marketplace pairing each client with a human tutor and a personalised LLM assistant for 24/7 support. Ran early customer discovery with parents, tutors and institutional stakeholders.

SKILLS, LANGUAGES & INTERESTS

Programming: Python (pandas, scikit-learn, NumPy), SQL, R, Git, Docker, Streamlit
Platforms: Google Cloud Platform (BigQuery), Databricks, Looker, Vercel, Supabase
Methods: Predictive modelling, imbalanced classification, recommender systems, offline evaluation & annotation studies, regression analysis, LLM evaluation, RAG, Vibe Coding
Languages: French (native), English (fluent), Japanese (5-month intensive course, Tokyo)
Interests: Basketball, entrepreneurship, classical music, East Asian culture, podcasts